SLEEPING MACHINES • TUTORIAL 4

Information Theory for Learning Signals

What forward information buys, Gaussian coding, and economical credit messages

Why this tutorial

A credit learner receives a forward packet and must predict how future errors affect a past computation. Information theory asks what uncertainty that packet removes and how many bits a useful credit message needs. It separates *predictable structure*, irreducible future noise, and communication cost. A coding bound is not a measured training-speed bound.

1 Entropy and usable side information

For a discrete variable X, $H(X) = -\sum_x p(x) \log p(x)$; logarithms here are natural, so units are nats. Conditional entropy averages uncertainty after observing Y. Mutual information is

$$I(X;Y) = H(X) - H(X \mid Y) = \mathbb{E} \log \frac{p(X,Y)}{p(X)p(Y)} \geq 0.$$

For continuous variables use differential entropy; it can be negative, while mutual information remains nonnegative. A deterministic statistic $Z = f(Y)$ obeys the data-processing inequality $I(X;Z) \leq I(X;Y)$: compression cannot create information about X.

What should the packet predict? Let G be future credit and Y the currently available packet. Under squared error, the optimal predictor is $m(Y) = \mathbb{E}[G \mid Y]$. For any predictor h,

$$\mathbb{E}\|G - h(Y)\|^2 = \mathbb{E}\|G - m(Y)\|^2 + \mathbb{E}\|m(Y) - h(Y)\|^2.$$

Derivation: expand the square and condition on Y; the cross term vanishes because $\mathbb{E}[G - m(Y) \mid Y] = 0$. More capacity can reduce the second term, but not unpredictable future noise. In a nonstationary stream both the conditional mean and its noise law may evolve.

2 The Gaussian channel and rate–distortion

For a real Gaussian $X \sim \mathcal{N}(0, v)$, $h(X) = \frac{1}{2}\log(2\pi e v)$. If $Y = X + N$, with independent $N \sim \mathcal{N}(0, n)$, then

$$I(X;Y) = \tfrac{1}{2}\log(1 + v/n), \qquad \mathrm{Var}(X \mid Y) = \frac{vn}{v+n}.$$

The mean-squared-error rate–distortion function of an i.i.d. real Gaussian source is

$$R(D) = \left[\tfrac{1}{2}\log(v/D)\right]_+, \qquad [x]_+ = \max(x, 0).$$

Lower bound: $I(X;\widehat{X}) = h(X) - h(X \mid \widehat{X}) \geq h(X) - h(X - \widehat{X})$, and Gaussian noise maximizes entropy at a fixed second moment D. Equality is attained by a Gaussian test channel. The operational theorem concerns long blocks and average distortion, not a one-message quantizer.

Key idea

For a *circular complex* Gaussian with $v = \mathbb{E}|X|^2$, the formula is $R(D) = [\log(v/D)]_+$ nats per complex sample. Its real and imaginary parts each have variance $v/2$. This factor of two matters for the complex temporal modes in our notes.

3 Weighted reverse water-filling

Suppose independent real Gaussian modes have variances v_i and credit errors have importance weights $w_i > 0$. Minimize the weighted distortion $\sum_i w_i D_i$ at a fixed total coding rate, with $0 < D_i \le v_i$. On active modes the Lagrangian derivative gives

$$w_i - \frac{\beta}{2D_i} = 0, \qquad D_i = \min(v_i, \theta/w_i), \quad \theta = \beta/2.$$

Only modes with $w_i v_i > \theta$ receive positive rate. The others can be omitted at the chosen distortion level. Weights may represent a producer sensitivity, but cross-mode correlations require a joint metric or an appropriate change of basis. A diagonal approximation must be labelled.

Example 1. Take $v = (9, 1)$, $w = (1, 4)$ and water level $\theta = 2$. Then $D = (2, 1/2)$; both modes are active. Rate is $\frac{1}{2}\log(9/2) + \frac{1}{2}\log 2 = \log 3$ nats, and weighted distortion is $2 + 4(1/2) = 4$.

4 Side information and noisy teachers

If (X, Y) are jointly Gaussian, write $X = \mathbb{E}[X \mid Y] + Z$ with an independent Gaussian innovation of variance v_c. When both ends know Y, encode only Z: $R_{X|Y}(D) = [\frac{1}{2}\log(v_c/D)]_+$. The Gaussian Wyner–Ziv theorem gives the same rate when only the decoder knows Y, under memoryless jointly Gaussian sources and squared error. It requires block coding; it is not a claim that any learned decoder attains the bound. For $D < v_c$, the rate saving is $\frac{1}{2}\log(v/v_c) = I(X;Y)$.

Indirect source coding. Suppose the desired signal is X, but the encoder only observes $Y = X + N$. Put $m(Y) = \mathbb{E}[X \mid Y]$, $d_0 = \mathbb{E}(X - m(Y))^2$ and $v_m = \mathrm{Var}(m(Y))$. Every reconstruction based on Y satisfies

$$\mathbb{E}(X - \widehat{X})^2 = d_0 + \mathbb{E}(m(Y) - \widehat{X})^2.$$

Thus $D < d_0$ is impossible, and the real Gaussian indirect rate is $[\frac{1}{2}\log(v_m/(D - d_0))]_+$ for $D > d_0$. Noisier gradient teachers need denoising or more information, not just a larger predictor.

Exercises

Exercise 1. With $v = 4$, $n = 1$, calculate the conditional variance and information.

Solution. $v_c = 4/5$ and $I = \frac{1}{2}\log 5$ nats. A real Gaussian code at $D = 0.2$ costs $\frac{1}{2}\log 4$ with this side information, versus $\frac{1}{2}\log 20$ without it.

Exercise 2. A mode has weighted variance $w_i v_i = 0.3$ and water level 0.5. What is sent?

Solution. Nothing: $D_i = v_i$ and its allocated rate is zero. This accepts distortion; it is not exact credit.

Exercise 3. Conditional residual variance falls by $e^{-2r\tau}$. Find the information gain under Gaussian assumptions.

Solution. The log variance ratio is $2r\tau$. Information is $r\tau$ nats per real mode, or $2r\tau$ per circular complex mode.

Where this appears in our work

Note 164 Part II uses complex Gaussian innovations, forward side information and weighted coding. Note 165 separates total capacity from active credit dimensions. The real online-credit prototype learns conditional state cotangents; sparse residual audits correct its errors. Gaussian coding identities do not certify that prototype's calibration or runtime. Basic.lean proves log-ratio and active-set algebra, not the Shannon or Wyner–Ziv coding theorems.

Further reading. MIT 6.441, chapters 23–25, rate–distortion lecture notes; Wyner and Ziv, *The rate-distortion function for source coding with side information at the decoder* (1976).